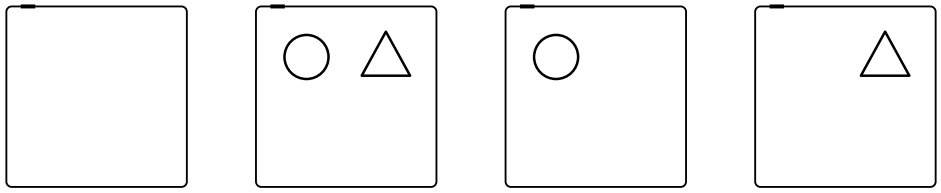
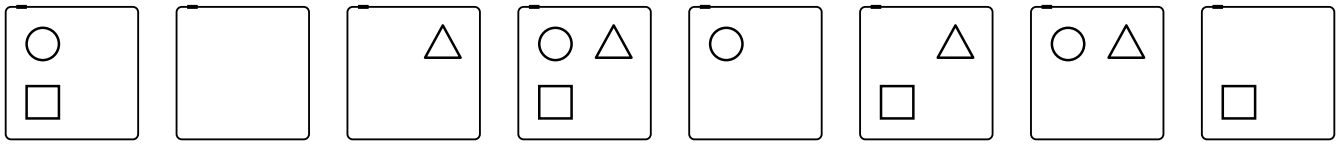
A card fits inside another when all its pictures are on the other card. The empty card fits inside every card. In a collection, no card may fit inside another. Use a card only once in a collection.

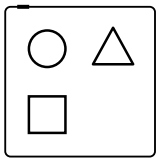
Problem 1: Find every collection of one or more cards from this deck that follows the rule. How do you know you have them all?

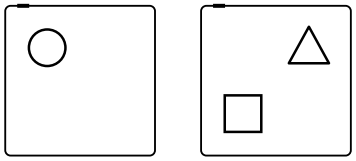


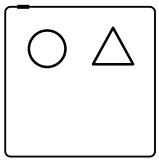
Problem 2: Find the largest collection you can from this deck. Find every collection of that size.



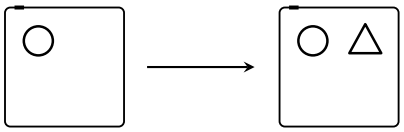
Problem 3: Use the same deck. Keep each starting collection and add as many cards as you can. Does being unable to add a card mean a collection is as large as possible?



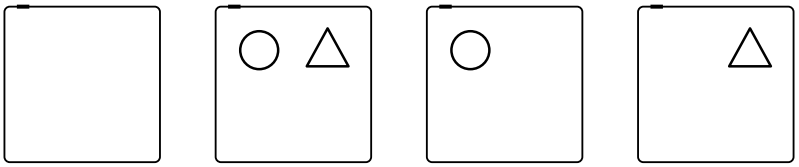


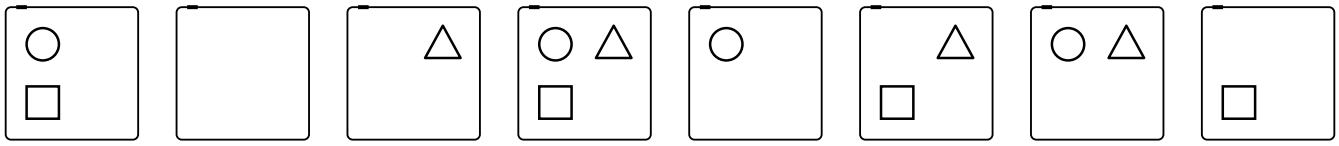


Problem 4: For each deck, arrange every card into exactly one row, with each card fitting inside the next card in its row. A card can be alone in a row. Make as few rows as possible.

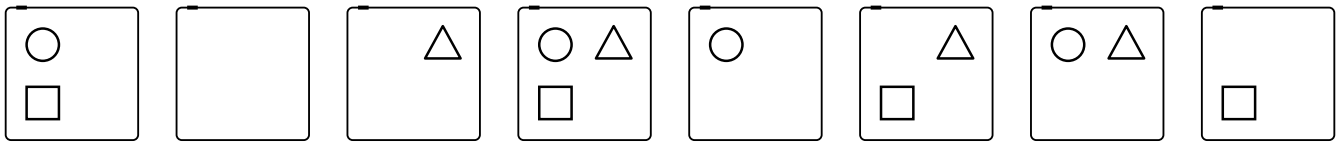


One nested row.

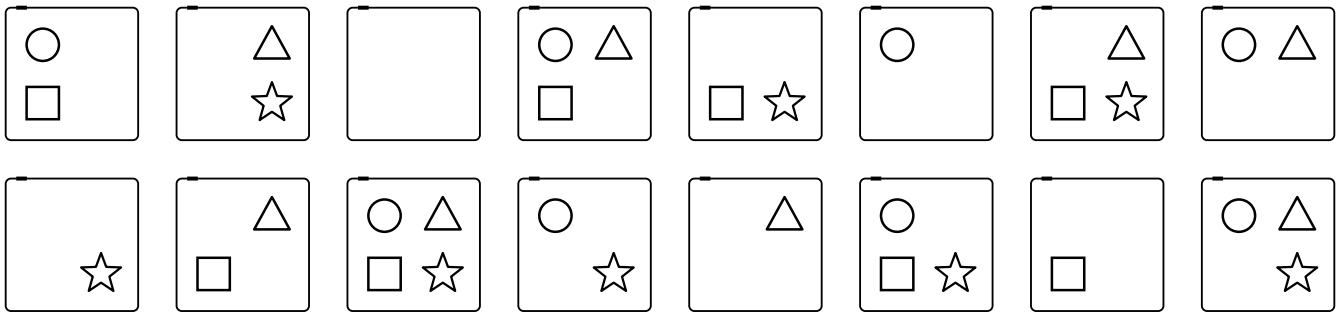




Problem 5: Could four cards from this deck follow the collection rule? Explain why your answer covers every choice of four cards.



Problem 6: Use this sixteen-card deck. Find the largest collection you can.



Problem 7: Arrange all sixteen cards from Problem 6 into as few rows as possible, with each card fitting inside the next card in its row. Every card must appear in exactly one row.

Problem 8: What is the largest possible collection from the sixteen-card deck? Explain why no larger collection can work.